

Guide to Importing WGCNA Networks into Cytoscape

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June 15, 2026

1 Introduction

This document outlines the step-by-step procedure for importing and visualizing the edge and node data tables generated from a Weighted Gene Co-expression Network Analysis (WGCNA) workflow into Cytoscape.

2 Step-by-Step Pipeline

1. Import the Network Structure (Edges)

- Launch Cytoscape and navigate to **File > Import > Network from File...**
- Select your edge file (e.g., `Edges_Tumor_Unsigned.txt`).
- In the configuration dialog, map the columns explicitly:
 - Set `gene1` (or `gene1.name`) as the **Source Node** (Green circle icon).
 - Set `gene2` (or `gene2.name`) as the **Target Node** (Red target icon).
 - Set `weight` as an **Edge Attribute** (Blue list icon).
- Click **OK** to render the unformatted network framework.

2. Import Attribute Metadata (Nodes)

- Navigate to **File > Import > Table from File...**
- Select your node file (e.g., `Nodes_Tumor_Unsigned.txt`).
- Under *Where to Import Table*, verify that **Import Data as: Node Table Columns** is selected.
- Ensure the **Key Column for Network** matches the primary identifier used during the network edge import step (`GeneID` or `GeneSymbol`).
- Click **OK** to link the metadata to the active network nodes.

3. Apply Force-Directed Physics Layout

- Go to the top menu bar and click on **Layout**.
- Select **Prefuse Force Directed Layout > (unweighted)** or **Edge-weighted Spring Embedded Layout** using your `weight` column attribute.
- This physically clusters tightly co-expressed genes into distinct spatial modules.

4. Map WGCNA Module Colors

- Navigate to the left-hand panel and click on the **Control Panel > Style** tab.
- Select the **Node** sub-panel at the bottom.
- Expand the **Fill Color** dropdown menu.
- Set the **Column** to `ModuleColor` and the **Mapping Type** to **Discrete Mapping**.
- Manually assign the literal visual color palette to each corresponding character string entry (e.g., set the hex/color value for the “turquoise” string to actual turquoise).

5. Configure Node Labels

- Within the same **Style > Node** panel, locate the **Label** property field.
- Set the target rendering **Column** to `GeneSymbol`.
- Set the **Mapping Type** to **Passthrough Mapping** to dynamically swap Ensembl identifiers for standardized gene names.

3 Optimization Tips

- **Density Filtering:** If your network layout is too dense to interpret, navigate to the **Filter** tab on the left control panel. Add an *Edge Table column filter* targeting the `weight` attribute and increment the threshold slider (e.g., ≥ 0.25) to isolate core intramodular hub structures.
- **Resolution Scaling:** Toggle **View > Graphics Details** to hide labels dynamically when zooming out, maximizing performance on large network matrices.